

GOVERNMENT FRAGMENTATION AND ECONOMIC GROWTH

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Abstract—We estimate the impact of local government fragmentation on economic activity in Indonesia from 2000 to 2014, when the number of districts increased by 50%. Exploiting idiosyncratic variation in the timing of district splits, we find that fragmentation reduces district GDP in the short term despite large increases in central transfers. The GDP decline is larger in “child” districts that acquire a new capital and government. Furthermore, splitting districts focus spending on administration without improving public services or reducing red tape and corruption. The downsides of fragmentation due to economies of scale and low bureaucratic capacity outweigh potential upsides.

1. Introduction

DECENTRALIZATION—the devolution of responsibilities to subnational governments—is a policy choice that carries significant implications for economic activity and welfare. Centralized systems exploit economies of scale in the provision of public services, are better positioned to internalize policy spillovers, and feature fewer, more capable politicians and bureaucrats. Decentralized systems can better accommodate diverse citizen preferences, incorporate information about local conditions, or promote accountability via interjurisdictional competition. While greater accountability or improved public service provision are valuable goals on their own, the stated aim of most decentralization policy is to enhance economic prosperity through these channels.¹ However, the net impact of decentralization on economic growth remains an open question.

This paper examines a prevalent feature of decentralization reforms across the developing world: the creation of new local governments. Similar to the transfer of responsibilities from central to local authorities, government frag-

mentation expands the number of jurisdictions responsible for policymaking, albeit while keeping the roles of different tiers of government unchanged. Consequently, many advantages and disadvantages of decentralization apply with equal force to the proliferation of local governments.²

We study an extraordinary period of government fragmentation in Indonesia between 2000 and 2014, during which the number of districts increased from 341 to 514 with the stated goal of accelerating regional economic development (Regulation No. 129/2000). Exploiting idiosyncratic variation in the timing of district splits generated by two national moratoria, we find that fragmentation reduces economic growth in the short run. Splitting districts experience a cumulative loss of 18% of GDP over five years following a split, compared to nonsplitting districts. We observe a similar decline when using household survey data on expenditures as well as nighttime luminosity data. This decline is remarkable considering that splits lead to a large increase in central transfers. Based on a conservative multiplier assumption, the fiscal shock alone should result in a 17% cumulative increase in GDP.

To examine whether the results mask general equilibrium effects, we compare the effects of doubling the number of districts at three levels of aggregation: original district (2000 borders), province-by-island, and province. We find that the negative growth effects are more pronounced at higher levels of geography. While these effects are not statistically different from one another, they rule out the possibility that splits simply reshuffle economic activity across space. District proliferation lowered nationwide GDP in Indonesia.

We consider several mechanisms through which local government fragmentation can affect growth. Theory motivated by developed country settings emphasizes the roles of tax competition and sorting by preferences over the level of public services.³ These mechanisms are unlikely to be important in our setting, as Indonesian districts are extremely limited in their ability to tax income and property, a common situation in the developing world (Gadenne & Singhal, 2014). However, districts do collect business licensing fees and receive shared revenue from business taxes administered by the central government. Therefore, government fragmentation could increase competition for mobile firms, limit rent seeking, and improve the business environment (Brennan

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¹See, for example, World Bank (1999), United Nations (2009), and International Monetary Fund (2009).

²A distinct but related phenomenon is political party fragmentation, for example, many jurisdictions controlled by different political parties. This type of fragmentation can also result in policymaking that neglects spillover effects (Magontier et al., 2022).

³Increasing the number of jurisdictions can facilitate sorting by taste (Tiebout, 1956), which can either increase growth (Brueckner, 1999, 2006) or decrease growth (Benabou, 1993, 1996), depending on the composition of the resulting heterogeneous communities. Government fragmentation can also increase competition for mobile capital, leading to lower capital tax rates and higher growth (Hoyt, 1991; Hatfield, 2015). See Agrawal et al. (2022) for a review of the literature on decentralization and growth.

& Buchanan, 1980; Fisman & Gatti, 2002; Arikan, 2004). District proliferation could also promote accountability via stiffer yardstick competition, in which voters judge the quality of local politicians by comparing their outcomes to those of similarly situated jurisdictions (Besley & Case, 1995).⁴ On the other hand, fragmentation could impede growth if there are economies of scale in the provision of public goods (Oates, 1972; Alesina & Spolaore, 1997), or if bureaucratic capacity falls because the new governments are staffed with less qualified civil servants.

We find that bureaucratic capacity and economies of scale are important mechanisms, but accountability is not. GDP falls both in parent districts, which retain the original government, and in newly created child districts, but the decline is larger in child districts.⁵ The difference in economic impact on parent and child districts mirrors the difference in quality of their civil servants, who are tasked with the majority of local public service provision. Child districts tend to have younger, less educated civil servants. The quality of the bureaucracy therefore matters but does not explain all of the decline. Economies of scale also play a role. Splitting districts devote a greater share of expenditure to administrative expenses and fail to improve public services—despite the increase in central transfers. Bureaucratic capacity also contributes to the disappointing results for public services. District splits lead to delays in the execution of physical capital spending plans, with only child districts struggling to implement plans following a split. By contrast, the hypothesized gains in accountability never materialize. Following a split, firms report paying bribes at higher rates, and there are no improvements in the regulatory environment. Unsurprisingly, firm productivity does not increase.

The baseline model nonparametrically controls for region-by-year effects, so that identification hinges on a conditional parallel trends assumption: within each region, GDP would have grown at the same rate in splitting and nonsplitting districts in the absence of splitting. This assumption could be violated if districts select into splitting based on recent economic shocks. However, institutional factors deprive districts of precise control over the timing of splits, lessening this concern. Furthermore, splitting and nonsplitting districts grew at the same rate, on average, prior to the split.

The conditional parallel trends assumption could also be violated if baseline characteristics that matter for growth are unbalanced for splitting and nonsplitting districts within a region. To address this concern, we show that the results are robust to controlling for time-varying effects of baseline ethnic fractionalization, urbanization, age structure, and education.

⁴However, in the model of Boffa et al. (2016), fragmentation reduces accountability due to heterogeneity in voter information and diminishing marginal benefits of information. The relationship between decentralization and accountability is ambiguous in the model of Bardhan and Mookherjee (2000).

⁵More precisely, GDP falls more in child districts relative to the expected path of GDP given the fiscal multiplier effect of splits.

A final, unrelated problem is that the timing of splits varies across districts. As a result, estimates based on a restrictive two-way fixed effects model may not recover a reasonably weighted causal effect if treatment effects are heterogeneous across time or districts. We circumvent this problem by estimating separate difference-in-differences estimands for each splitting cohort and time period. We then report the weighted average of the cohort-specific estimates.⁶

Our setting provides several advantages. First, the massive scale of the reforms taking place in the world's fourth-largest country makes this an important episode to study. If decentralization and fragmentation are to be recommended as a rule, it is necessary to understand what effect these policies can have at a wide scale. Second, two national moratoria on splitting generated idiosyncratic variation in the timing of splits, strengthening our difference-in-differences design. Much of the earlier work on decentralization and growth relies on cross-country comparisons across very different contexts and with confounding events around decentralization episodes. Finally, Indonesia is one of the few developing countries that provides a consistent GDP series at the second subnational level. Research on decentralization and local development in other settings has relied on imperfect proxies such as nighttime luminosity.

Our results are informative for the policy debate over administrative unit redistricting. Local governments have proliferated in many parts of the developing world and eastern Europe with the goal of promoting economic development (Swianiewicz, 2010; Grossman & Lewis, 2014; Grossman et al., 2017).⁷ Our analysis shows that the disruption caused by the creation of new governments can more than offset the fiscal benefits enjoyed by splitting units, reducing growth in the short run. To avoid the pitfalls of fragmentation, policymakers should pay special attention to the quality of the new government personnel and consider how government scale will affect public good provision.

This paper contributes to multiple literatures. First, it contributes to the literature on administrative unit redistricting. This literature focuses on public expenditure and service delivery, not growth (see Gendźwiłł et al., 2021, for a review). However, there are two recent exceptions. Dahis and Sberman (forthcoming) find that municipal splits in Brazil improve public service provision and increase nighttime luminosity in new municipalities, but have no impact on private sector employment. By contrast, Cohen (2024) finds that district splits in Uganda harm the quality of infrastructure and reduce nighttime luminosity in new districts. Prior research on district splitting in Indonesia has focused on public services (Lewis, 2017; Singhania, 2022), ethnic conflict (Bazzi & Gudgeon, 2021), and deforestation (Burgess et al., 2012; Alesina et al., 2019).⁸ In contrast to these papers, we

⁶Our approach draws on the estimators of Wooldridge (2025) and Sun and Abraham (2021).

⁷Notable examples include Brazil, Nigeria, and Vietnam.

⁸Our findings on public services are consistent with those of Lewis (2017), who finds that splits do not improve public service delivery. We

focus on economic growth. Related work in political science studies the political motivations behind administrative unit splitting in the developing world (Grossman & Lewis, 2014; Pierskalla, 2016). Research in developed country settings focuses on the effect of amalgamations on public expenditure.⁹

Second, our paper contributes to the literature on decentralization and growth. Cross-country analyses are challenging due to endogeneity concerns and the difficulty of finding a single measure of decentralization that summarizes inter-governmental relations and can be consistently measured for all countries (Oates, 1993; Rodden, 2004). In response to these challenges, recent work uses within-country variation in the number of local governments in contexts very different from our own. Exploiting cross-sectional variation in the number of local governments in a metropolitan area, Stansel (2005) and Hatfield and Kosec (2013) find that fragmentation increases growth in the United States. Similarly, Zhang et al. (2019) find that fragmentation increases local growth in China, but only when starting from a low level of fragmentation. In those studies, the local governments had been operating for a significant amount of time, whereas we study the creation of new governments.

Finally, our paper is related to research documenting the downsides of decentralization in practice in developing countries. Lipscomb and Mobarak (2017) find that decentralization exacerbates water pollution externalities in Brazil. Also examining Brazil, Kresch (2020) shows that uncertainty over the responsibilities of different tiers of government can deter public investment. Our paper, by contrast, underscores the importance of bureaucratic capacity and scale for realizing the potential gains from fragmentation.

In section II, we describe the institutional context of Indonesia's decentralization reforms. Section III describes our data set, which combines data on districts, households, villages, and manufacturing firms. Section IV then explains the empirical strategy, and section V presents the results. In section VI, we examine potential mechanisms behind the results. Section VII compares our results to those of previous studies and discusses potential implications for long-run growth. Section VIII provides concluding remarks.

II. Empirical Context

A. Indonesia's Decentralization Reforms

Following the resignation of autocratic ruler Suharto in 1998, Indonesia transitioned to democracy and instituted a series of political and fiscal reforms. Indonesia is currently divided into 34 provinces, the first tier of subnational government, which mostly play a coordinating role. Districts, the

second tier of subnational government, are categorized as either urban districts (*kota*) or rural districts (*kabupaten*); however, political institutions and fiscal responsibilities are the same for both types. Subdistricts (*kecamatan*) and villages are the third and fourth tiers of government, respectively.

Shortly after democratizing, Indonesia implemented the “Big Bang” decentralization reforms, which expanded the authority of district governments (Shah et al., 2012). From 2001 to 2014, districts were responsible for the provision of education, health, environmental, and infrastructure services (Law No. 22, 1999a), with 30% of consolidated government expenditure occurring at the district level (World Bank, 2003).¹⁰ In practice, infrastructure projects involved collaboration between districts and villages: districts financed upgrades and procured engineers while villages were responsible for maintenance (World Bank, 2010). Overall, districts determined the vast majority of policies on local public services.

The Village Law (Law No. 6, 2014) ushered in the second phase of decentralization starting in 2015, devolving some responsibilities from districts to villages. We therefore limit our sample to the years 2001 to 2014 to hold constant the responsibilities of subnational governments.¹¹

Central transfers financed the vast majority of district expenditure over this period, as specified in Law No. 25 (1999b). The largest transfer is an unconditional, nonmatching grant known as the General Allocation Grant (*Dana Alokasi Umum*), or “general grant” for short. On average over half of district revenue comes from this grant. Half of the grant funds the so-called basic allocation, which covers the civil service wage bill. The rest is determined according to a fiscal gap formula that uses proxies for expenditure needs (e.g., population, land area, poverty) and fiscal capacity (e.g., predicted revenue from other sources) (World Bank, 2007; Cassidy, 2025). The central government regulates recruiting and employment contracts to prevent excessive spending on civil servant wages (Shah et al., 2012). In all other areas, district spending out of the general grant is essentially unconstrained.

Some districts receive significant Shared Natural Resource Revenue (*Dana Bagi Hasil Sumber Daya Alam*), which is distributed on the basis of local natural resource extraction and is unconditional. Finally, a small portion of the district budget (around 6%) comes from Special Allocation Grants (*Dana Alokasi Khusus*), which are conditional, matching grants awarded on a discretionary basis. Law No. 33 (2004) modified the grant allocation formulas and increased the budget for the general grant (see Cassidy, 2025).

While Indonesia substantially decentralized expenditure responsibilities, it did not devolve tax authority to a similar degree. The central government sets tax rates on sales, individual income, and corporate income, and it administers

build on their results by using a longer time horizon and a larger set of indicators of public service provision. Singhanian (2022) focuses on the interaction effect between splitting and direct mayoral elections, using a more limited set of years.

⁹See, for example, Reingewertz (2012), Breuille and Zanaj (2013), and Blom-Hansen et al. (2016). Erlingsson et al. (2021) provide a rare study of municipal splitting in a developed country setting.

¹⁰See United Nations (1999) for an English translation of Law No. 22 (1999a).

¹¹Technically, our sample includes years prior to 2001 to aid in the estimation of pretrends. These years do not enter into the estimates of treatment effects.

these taxes. The central government also set property tax rates until 2010. Between 2011 and 2014, districts were allowed to voluntarily adopt tax-setting authority over the property tax, with the vast majority of districts acquiring this authority in 2014. Because of the timing of this reform and the fact that districts were reluctant to deviate from predcentralization property tax rates (von Haldenwang, 2017), the property tax was not an important local policy tool during the study period.

The central government returns a portion of the revenue to the district where the taxes were collected. The sharing rate is 9% for the property tax, 16% for the property transfer tax, and 12% for the income taxes. Minor local taxes and user fees, such as the motor vehicle tax and the hotel tax, are under the purview of districts but are subject to a rate ceiling set by the center. Almost all districts charge the maximum rate permitted (World Bank, 2003). Districts are also responsible for many business licenses, which formally may be obtained by paying a fee set by the district.

Prior to democratization, President Suharto appointed district heads. After the country transitioned to democracy, district voters directly elected members of the local parliaments, and these parliaments selected the district heads. Starting in 2005, districts introduced direct elections of district heads in a staggered fashion. The election timing was staggered because incumbents were allowed to finish their five-year terms, which for idiosyncratic reasons were not synchronized across districts.

B. District Fragmentation

Local elites typically spearhead the effort to create new districts. Formally, district parliamentarians petition to split a district, with the approval of the mayor of the original district, through a process known as *pemekaran*, or “blossoming.” The central government then decides whether to accept or reject the petition based on technical criteria. At the outset, the proposed parent and child districts all had to contain at least three subdistricts and possess sufficient economic capacity (Regulation No. 129, 2000).¹² The new district boundaries follow the borders of existing subdistricts. Upon approval, the Ministry of Home Affairs appoints an interim head of the child district; the interim legislature features the same party composition as that of the original district. One to two years later, the first elections take place in the child district (Fitriani et al., 2005).

During the sample period, the number of districts increased from 341 to 514, an increase of 50%. Figure 1a shows that the growth of district governments over this period was interrupted twice—from 2004 to 2006 and from 2009 to 2012—by national moratoria, creating exogenous variation in the timing of districts splits. Over a hundred districts had applied to split before the first moratorium was announced and did not know when their application might

be approved. After the first moratorium, the central government strengthened the splitting regulations to require that (a) the proposed districts all contain at least four subdistricts (in the case of *kota*) or five subdistricts (in the case of *kabupaten*); and (b) the original district has existed for at least seven years (Regulation No. 78, 2007). We further discuss the moratoria in the context of our identification strategy in section IV.

Figure 1b provides a map of district borders in 2000 (thick black lines) and 2012 (thin gray lines), with districts that split over this period shaded in purple.¹³ About one-third of the original districts split at least once between 2001 and 2014. The map shows that the island of Java, the historical center of economic and political power, has relatively few districts that split. By contrast, district splitting was widespread in the “outer islands” of Sumatra, Kalimantan, Sulawesi, Maluku, Papua, and Nusa Tenggara.

There are several motivations for district creation. The first is purely political. The creation of a new government, with no entrenched incumbents, provides an opportunity for local elites to gain political power. Second, redistricting may satisfy local desires for more ethnically homogeneous jurisdictions (Fitriani et al., 2005; Pierskalla, 2016). Finally, there is a strong fiscal incentive to split. The general grant essentially has a large fixed component that does not depend on population. This virtually guarantees that transfers increase in per capita terms in both the child and parent districts following a split.

Bazzi and Gudgeon (2021) document that splitting causes total fiscal transfers to increase by 20% on average, while Cassidy (2025) shows that district expenditure increases roughly one-for-one with the general grant. Therefore, in the absence of other effects from fragmentation, splits should increase local GDP via a fiscal multiplier effect. In the empirical analysis, we compare the actual GDP response to splitting to a benchmark implied by the increase in transfers and local fiscal multipliers estimated in the literature.

III. Data

This section describes key features of our data and outcome variables. Further details on data sources and variable construction can be found in online appendix A. All outcome variables are aggregated to the level of district borders in 2000 unless specified otherwise. We identify splits using administrative data on district establishment dates, and we aggregate variables to original borders using a crosswalk provided by the World Bank.

A. District GDP

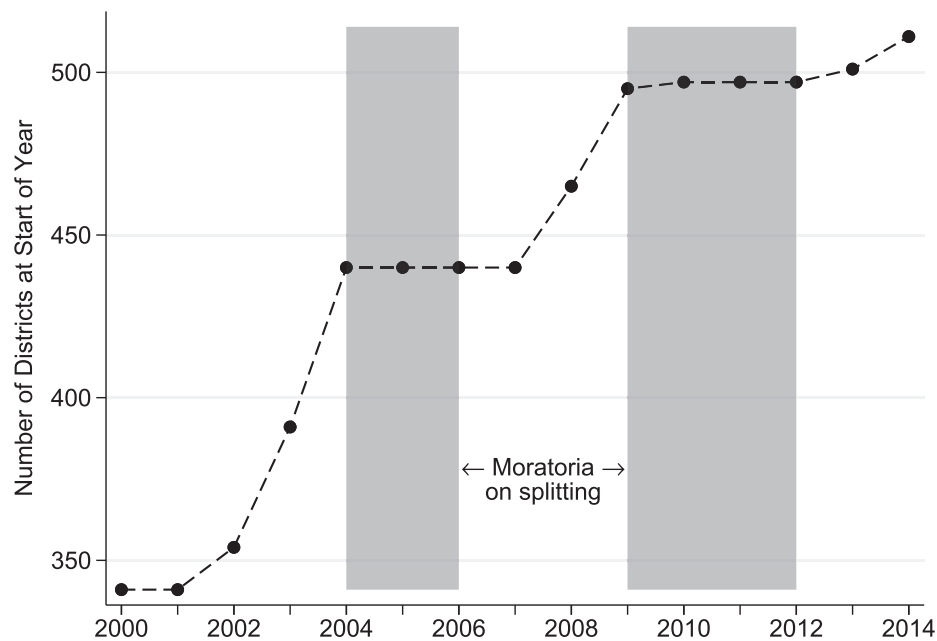
Our main outcome variable is district-level GDP. The Central Bureau of Statistics (Badan Pusat Statistik, BPS) produces estimates of district GDP, which are disseminated

¹²Districts are also allowed to amalgamate, but this has not occurred in practice.

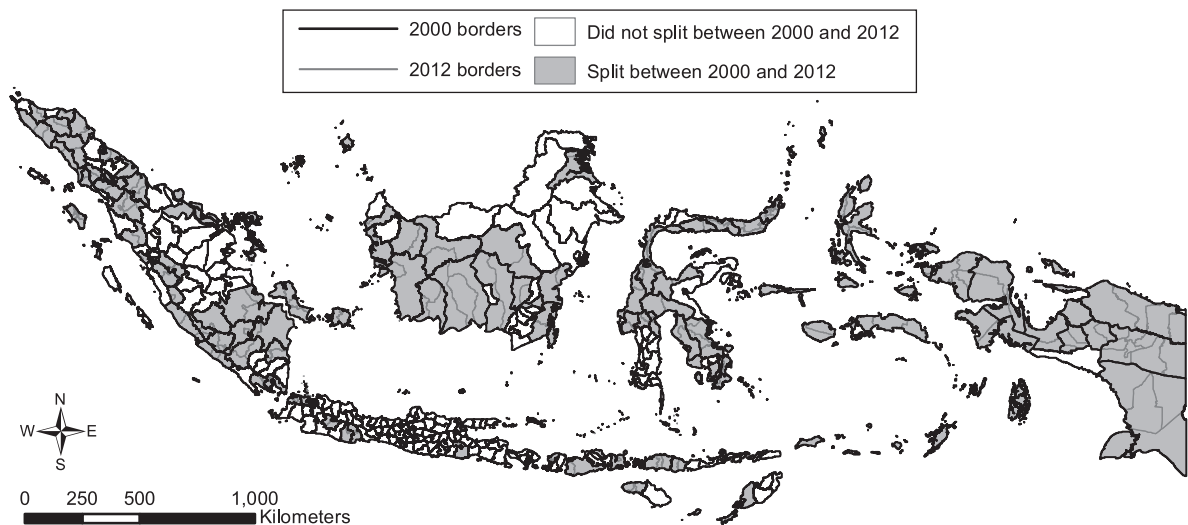
¹³As discussed below, we omit districts that split for the first time during the period 2013–2014.

FIGURE 1.—INDONESIA’S DISTRICT PROLIFERATION ACROSS TIME AND SPACE

(a) Two Moratoria Generated Idiosyncratic Variation in the Timing of District Creation



(b) District Splitting Was Geographically Widespread



(a) Number of districts at the start of the year. The two flat portions of the graph are due to moratoria on district creation imposed by the central government during the periods 2004–2006 and 2009–2012. (b) District borders in 2000 and 2012 based on the 2012 district shapefile provided by the Central Bureau of Statistics and the district crosswalk provided by the World Bank’s Indonesia Database for Policy and Economic Research.

by the World Bank’s Indonesia Database for Policy and Economic Research (INDO-DAPOER). The GDP series spans 2000 to 2013. BPS often reports separate GDP estimates for parent and child districts in selected years prior to a split. This allows us to estimate the growth impacts of splitting separately for parent and child districts. For districts that experienced multiple splits, we aggregate parent and child GDP up to district borders as of the year of the first split, because this is the level at which GDP is usually reported. (See online appendix A for details.)

B. Bureaucratic Capacity

We measure bureaucratic capacity using data on civil servants from multiple sources. The National Civil Service Agency (Badan Kepegawaian Nasional, BKN) provides data on the number of civil servants in each district from 2005 to 2015.¹⁴ We also use data on education and age from the 2005 Intercensal Population Survey (Survei Penduduk Antar

¹⁴We are grateful to Jan Pierskalla for sharing these data with us.

Sensus, SUPAS) and the Indonesian National Labor Force Surveys (Survei Angkatan Kerja Nasional, SAKERNAS) of 2007 to 2009 and 2011 to 2015. We augment the data on civil servants with information on the education level of the first democratically appointed mayor from Mukherjee (2024).¹⁵

C. Firm Outcomes

We analyze firm-level outcomes using data from the Indonesian manufacturing survey of large and medium-sized firms (Survei Industri Besar/Sedang, IBS), which covers the universe of manufacturing establishments with at least twenty workers. We combine the surveys to construct an unbalanced panel of establishments over the period 1995 to 2014. For simplicity, we use the terms *firm*, *establishment*, and *plant* interchangeably, even though we cannot link establishments belonging to the same firm.

IBS contains information on outcomes such as the total value of production, number of employees, and industry of operation. Establishments also report their total indirect tax payments, which combine taxes and fees administered by districts (establishment license fees, building and land taxes, compulsory donations to finance local infrastructure) and taxes administered by the central government (sales taxes, import duties, excise taxes). Another outcome of interest is “gifts” paid by the firm to external parties. We interpret this variable as including bribe payments to officials, following Henderson & Kuncoro (2004, 2011).

To support our interpretation of gifts as bribes, in the appendix we show that the within-firm incidence of bribery over time is positively correlated with firms’ activities that require permits or licenses from the local government, such as electricity connection from the government, exports, land contracts, and building construction. By contrast, the incidence of bribery is negatively correlated with own generation of electricity or purchase of electricity from nongovernmental sources (appendix table B1).

D. Sample Selection

We drop all five districts in the province of Jakarta, as these districts are managed by the province. We also omit the five districts that split for the first time during 2013–2014, because we cannot measure four-year growth post-split for these districts, as the GDP series ends in 2013. The resulting sample contains 331 districts at the level of 2000 borders.

E. Summary Statistics

Appendix table B2 summarizes the baseline characteristics of splitting and nonsplitting districts. Of the 331 districts in the sample, 98 experienced at least one split during the sample period, while 233 never split. Initial

GDP was 8 log points lower in splitting districts than in nonsplitting districts; however, this difference is statistically insignificant. Splitting districts received more general grant revenue, were more ethnically fractionalized, and were less urbanized at baseline. While the ethnic fractionalization distributions exhibit excellent overlap for splitting and nonsplitting districts, the urbanization rate distributions do not (appendix figure B1). Many nonsplitting districts have urbanization rates in excess of 75%, whereas the urbanization rate is below 75% for all splitting districts. Splitting districts also tend to have younger and less educated populations, though these differences are smaller than those for ethnic fractionalization and urbanization.¹⁶

IV. Empirical Strategy

We use a difference-in-differences (DiD) strategy to estimate the effect of government fragmentation on economic growth. District creation is a political process. The decision of whether to split is correlated with several district-level characteristics, such as ethnic fractionalization. (See Pier-skalla, 2016, and appendix table B2.) Rather than relying on cross-sectional variation in whether a district ever split, our identification strategy exploits idiosyncratic variation in the timing of splits. This variation comes from two sources.

First, there is generally a multiyear lag between when a district applies for a split and when the central government approves the split, and there is considerable uncertainty over whether the split will be approved. Therefore, prospective leaders of a new district lack precise control over the timing of its creation.

Second, the national government imposed moratoria on district splitting from 2004 to 2006 and from 2009 to 2012, generating additional idiosyncratic variation in the timing of splits. A district that applied to split in 2002 likely would have been approved in 2007 rather than in 2004 as a result of the first moratorium. In fact, 114 district applications awaited consideration by the end of the first moratorium (Badan Perencanaan Pembangunan Nasional, 2007). The central government justified the moratoria by citing the fiscal costs of district creation—total intergovernmental grants increase whenever a new district is formed—and the lack of qualified personnel needed to run the new governments (Bazzi & Gudgeon, 2021). To the best of our knowledge, the central government never cited the local economic impacts of splitting as a reason for the moratoria.

After the split is approved, an interim government is appointed. One to two years later, a democratically elected government takes over, and the district starts receiving fiscal transfers from the central government. We define the split year as the first full calendar year following the passage of the legislation creating the new district(s). In this way, we include the transition process as part of the treated period.

¹⁵We are grateful to Priya Mukherjee for sharing these data with us.

¹⁶Appendix table B3 provides additional summary statistics, and appendix table B4 provides details on sample coverage.

Districts are defined according to the original district boundaries in 2000.

Because the timing of the first split varies across districts, a restrictive two-way fixed effects model that includes a single “post-split” dummy may not recover a reasonably weighted average treatment effect in the presence of treatment effect heterogeneity.¹⁷ To avoid this problem, we construct separate DiD estimators for each splitting cohort, defined according to the year of the district’s first split.¹⁸ We then take a weighted average of the cohort-specific estimates. The control group is always the set of districts that never split during the analysis period.

We nonparametrically control for region effects, so that identification relies on a conditional parallel trends assumption: splitting and nonsplitting districts located in the same region would have experienced the same GDP growth on average in the absence of the split. This assumption is weaker than the usual (unconditional) parallel trends assumption. In particular it allows for regional heterogeneity in growth trajectories that may be correlated with the incidence of splitting.

On a practical note, we scale changes in GDP by initial GDP rather than using changes in log GDP, to remain consistent with the fiscal multiplier literature. This allows us to relate percentage changes in GDP to changes in grants as a percentage of GDP. The results are very similar if we instead use changes in log GDP.

Define the set of splitting cohorts $\mathcal{E} \equiv \{2002, 2003, 2004, 2008, 2009\}$. Let $Y_{d,t}$ be real GDP in district d in year t , let E_d be the cohort of district d (with $E_d = \infty$ for never-splitters), and let $\mathbf{X}_d$ be a vector of region dummies.¹⁹ Define the cohort indicators $D_d(e) \equiv 1(E_d = e)$. We implement our estimator in two steps. First, we estimate the regression

$$\frac{Y_{d,e+h} - Y_{d,e-1}}{Y_{d,e-1}} = \alpha_{e,h} + \beta_{e,h} D_d(e) + \mathbf{X}_d' \boldsymbol{\lambda}_{e,h} + (D_d(e) \cdot \dot{\mathbf{X}}_d(e))' \boldsymbol{\gamma}_{e,h} + \varepsilon_{d,e,h} \quad (1)$$

for each cohort e and time horizon h , using the subsample of districts that either split for the first time in year e or never split. The variable $\dot{\mathbf{X}}_d(e)$ is defined as $\mathbf{X}_d - E(\mathbf{X}_d | E_d = e)$. The presence of the interaction term allows the covariates to have a different effect on potential outcomes in the treated and untreated states. Centering the covariates in the interaction term about their cohort-specific means gives $\beta_{e,h}$ an interpretation as the average treatment effect for cohort e . Under the conditional parallel trends assumption and a “no an-

ticipation” assumption, the OLS estimator $\hat{\beta}_{e,h}$ is consistent for the cohort-specific average treatment effect on the treated (CATT) (Sun & Abraham, 2021; Wooldridge, 2025).²⁰

Note that because the covariates are binary and the regression is fully saturated, the model does not impose any functional form assumption on how potential outcomes depend on covariates; the baseline model thus controls for region-by-year effects nonparametrically. In robustness checks we will add nonbinary covariates to equation (1), thereby imposing a linear relationship between the covariates and potential outcomes, while allowing this relationship to differ in the treated and untreated states and over time.

Our approach to estimating each $\beta_{e,h}$ is similar to the one proposed by Wooldridge (2025), with a few differences. Wooldridge (2025) uses the average outcome across all pretreatment years as the baseline. To remain consistent with the fiscal multiplier literature, we use the outcome in the year prior to the split as the baseline. Measuring the change in GDP relative to this baseline necessitates the use of cohort-specific equations as in equation (1). We stack these equations and jointly estimate the parameters. The other main difference is that Wooldridge (2025) forms counterfactual outcomes by averaging across all units that have not yet adopted the treatment as of year $e + h$. We instead use never-treated units as controls to ensure that the composition of the control group does not change over time.

In the second step, we combine the DiD estimates to form an estimate of the cohort-size-weighted average treatment effect on the treated,

$$\hat{\beta}_h = \sum_{e \in \mathcal{E}} \omega_e \hat{\beta}_{e,h}, \quad (2)$$

where ω_e is the share of splitting districts that belong to cohort e . We calculate standard errors for $\hat{\beta}_h$ that are robust to heteroskedasticity and clustering within original district boundaries.²¹ While the conditional parallel trends assumption is not testable, it would be more plausible if splitting and

²⁰Formally, let $Y_t(e)$ denote the potential outcome in year t if the district split for the first time in year e , with $e = \infty$ if the district never split, and let $\mathbf{X}$ be a vector of region dummies. The conditional parallel trends assumption states that

$$E \left(\frac{Y_{e+h}(\infty) - Y_{e-1}(\infty)}{Y_{e-1}(\infty)} \mid \mathbf{X}, E = e \right) = E \left(\frac{Y_{e+h}(\infty) - Y_{e-1}(\infty)}{Y_{e-1}(\infty)} \mid \mathbf{X}, E = \infty \right)$$

for all e and $h > 0$. Under this assumption and the “no anticipation” assumption, $Y_t(e) = Y_t(\infty)$ for $t < e$, each DiD estimator identifies a cohort-specific average treatment effect on the treated (Sun & Abraham, 2021; Wooldridge, 2025), that is,

$$\beta_{e,h} = E \left(\frac{Y_{e+h}(e) - Y_{e+h}(\infty)}{Y_{e-1}(\infty)} \mid E = e \right).$$

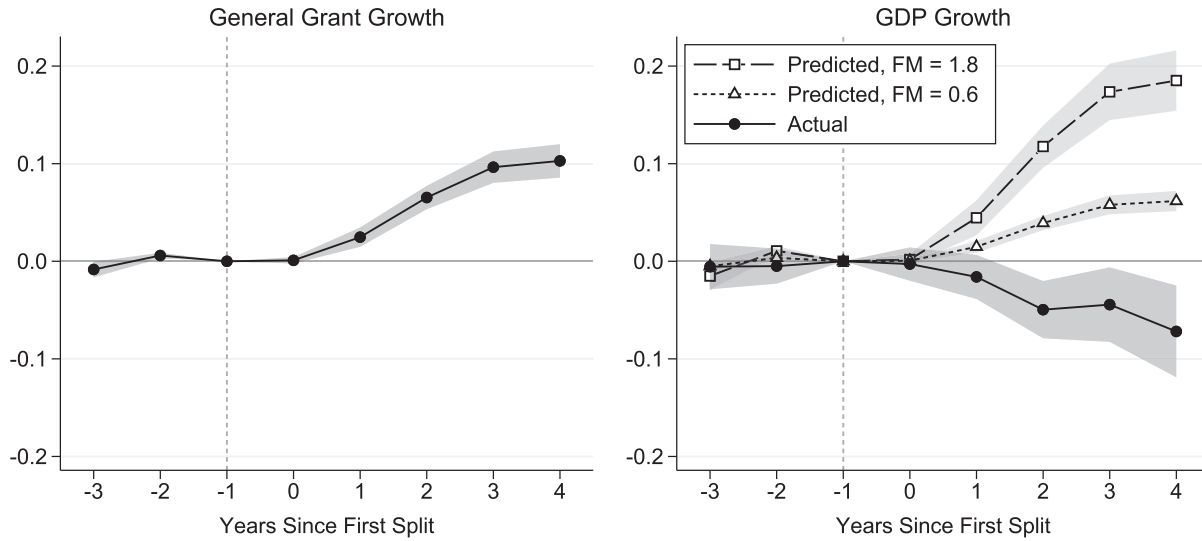
²¹Let $\hat{\mathbf{V}}_h$ be the cluster-robust estimator of the asymptotic variance-covariance matrix of the vector of estimators $\{\hat{\beta}_{e,h}\}_{e \in \mathcal{E}}$. Our estimator for

¹⁷See, for example, de Chaisemartin and D’Haultfœuille (2020), Goodman-Bacon (2021), Sun and Abraham (2021), Callaway and Sant’Anna (2021), and Borusyak et al. (2024).

¹⁸Only one district (Kabupaten Bengkalis) split for the first time in 2010. To avoid estimating a cohort-specific treatment effect based on a single treated unit, we code this district as having split in 2009. Bengkalis’s legislation creating a new district was passed on January 16, 2009, so it is basically accurate to code 2009 as the first full calendar year following passage of the legislation.

¹⁹Following the Central Bureau of Statistics, we code seven regions: Sumatra, Java, Nusa Tenggara, Kalimantan, Sulawesi, Maluku, and Papua.

FIGURE 2.—THE EFFECT OF DISTRICT SPLIT ON GENERAL GRANT AND GDP



This figure plots estimates of the cohort-size-weighted CATT (equation (2)) and their 95% confidence intervals. (The estimates for $h = -3$ omit the 2002 cohort and adjust the weights accordingly, because the data start in 2000.) The left panel shows the impact of the first district split on growth in general grant revenue relative to the year before the split, scaled by GDP in that year. The right panel shows the impact on GDP growth relative to the year before the split as predicted by fiscal multiplier values of 0.6 and 1.8 given the one-for-one increase in expenditure due to the increase in general grants. It also plots the impact on actual GDP growth. The confidence intervals are robust to heteroskedasticity and clustering by district.

nonsplitting districts in the same region experienced similar GDP growth prior to the split. In the next section, we test the hypothesis of common pretrends: $\beta_h = 0$ for $h < -1$.

Our estimation strategy has two advantages over the dynamic panel models commonly employed in the empirical growth literature. First, our approach is robust to treatment-effect heterogeneity across districts, whereas dynamic panel models assume homogeneous effects. Second, while dynamic panel models impose a specific functional form for dynamic treatment effects, our model makes no assumptions about dynamics.²²

V. Results

A. Impact of Fragmentation on Growth

Figure 2 shows that district splitting reduces district GDP, and the effect becomes more pronounced over time. This is noteworthy because central transfers increase within the original district boundaries after a split, which should positively affect GDP through increased government spending. The figure shows that general grant revenue eventually rises by 10% of pre-split GDP three years after the split. This increase in transfers resulted in a commensurate increase in district expenditure, as we show in section VIA. We multi-

ply this grant increase by the lowest estimate and a standard estimate of the local fiscal multipliers found in the literature (0.6 and 1.8) to generate predicted GDP responses to splitting.²³

If district splits only affected GDP through the fiscal multiplier channel, then GDP should have increased by 6% to 18% four years after the split. However, in reality, GDP begins to decline two years after the split and eventually falls by 7% after four years. This implies that splits adversely affect economic activity through nonfiscal channels. The general grant and GDP both exhibit parallel trends for splitting and nonsplitting districts prior to the split.

To summarize the impact of splitting in a single number, we estimate the cumulative effects on GDP. The first row of estimates in table 1 reports the cumulative impacts from the year of the split to four years after the split, $\sum_{h=0}^4 \hat{\beta}_h$. Districts experience a mechanical increase in general grant revenue as a share of pre-split GDP, adding up to 29.0% (S.E. = 2.3%) at the end of four years after the split (column 1). This implies a cumulative GDP increase of around 17% for a multiplier of 0.6 (column 2) and an increase of over 50% for a multiplier of 1.8 (column 3). In contrast, actual GDP experiences a cumulative decline of 18.5% (S.E. = 7.0%) (column 4). While the point estimate is substantial, the degree of uncertainty surrounding the estimate is not trivial; the 95% confidence interval includes cumulative declines as small as 5% and as large as 32%. Nevertheless, we can confidently reject positive impacts.

Given that the increase in the general grant should have resulted in positive growth, it is unlikely that the adverse impact on growth occurred through fiscal channels alone.

the asymptotic variance of $\hat{\beta}_h$ is $\omega \hat{V}_h \omega$, where ω is the vector of cohort shares $\{\omega_e\}_{e \in \mathcal{E}}$.

²²For example, consider the equation $y_{d,t} = \rho y_{d,t-1} + \beta D_{d,t} + \alpha_d + \gamma_t + \varepsilon_{d,t}$, where $y_{d,t}$ is log GDP, $D_{d,t}$ is a “post-split” dummy, and $\rho \in (0, 1)$. Iterating, we have $y_{d,t} = \rho^{h+1} y_{d,t-h-1} + \sum_{k=0}^h \rho^k (\beta D_{d,t-k} + \alpha_d + \gamma_{t-k} + \varepsilon_{d,t-k})$. Now this equation, combined with a sequential exogeneity assumption, implies that the causal effect of splitting on (log) GDP h years later is $\beta \sum_{k=0}^h \rho^k$. Hence, the treatment effect is monotonic in h and approaches $\beta/(1 - \rho)$ as $h \rightarrow \infty$.

²³Chodorow-Reich (2019) reviews the literature on local multipliers and finds a cross-study mean of 1.8.

TABLE 1.—GOVERNMENT FRAGMENTATION AND ECONOMIC GROWTH

	General Grant	Predicted GDP		Actual GDP
	(1)	(2) Multiplier = 0.6	(3) Multiplier = 1.8	(4)
Cumulative Effect of Split:				
All Splits	0.290*** (0.023)	0.174*** (0.014)	0.523*** (0.041)	− 0.185*** (0.070)
Early Splits	0.246*** (0.020)	0.148*** (0.012)	0.444*** (0.035)	− 0.130 (0.086)
Late Splits	0.412*** (0.066)	0.247*** (0.040)	0.742*** (0.120)	− 0.336*** (0.116)
<i>p</i> -value, H_0 : Early = Late	0.017	0.017	0.017	0.157
Observations	1,263	1,263	1,263	1,263
Districts	331	331	331	331

The first row reports estimates of the cumulative effect of the first district split, $\sum_{h=0}^4 \beta_h$, based on the cohort-size-weighted CATT estimator, equation (2). The second and third rows report separate cohort-size-weighted CATT estimates for early (2002–2004) and late (2008–2009) splits. The outcome is measured as the cumulative growth from the year prior to the split to four years after the split, relative to GDP in the year prior to the split. Standard errors, reported in parentheses, are robust to heteroskedasticity and clustering by district. * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

Evidence on local fiscal multipliers in developing countries is scarce, but the multiplier is never negative. At the national level, Kraay (2012, 2014) estimate a fiscal multiplier of around 0.4 in developing countries in the short-term in response to a windfall increase in expenditure similar to these central government transfers.²⁴ Subnational fiscal multipliers are typically higher than this in developed countries (Chodorow-Reich, 2019). For example, Serrato and Wingen-der (2016) estimate a local income multiplier of 1.7 to 2 in the United States. Estimates vary widely in developing country contexts. On the high end, Corbi et al. (2019) find that grants to Brazilian municipalities have a local income multiplier of around 2. On the low end, Guo et al. (2016) estimate a multiplier of 0.6 in China.

Even with the most conservative estimate of the fiscal multiplier at 0.6, the increase in the general grant should have significantly boosted GDP. However, our findings show that splits have a significant and negative effect on growth. This suggests that other factors may be contributing to the negative impact on growth, such as bureaucratic capacity and economies of scale in the provision of public goods. We explore mechanisms in section VI.

B. Robustness Checks

Baseline imbalance. A potential threat to our strategy is that several baseline covariates are unbalanced for splitting and nonsplitting districts (appendix table B2). Covariate imbalance could lead to a violation of the conditional parallel trends assumption if these covariates are related to trends in potential outcomes. For instance, ethnic fractionalization and urbanization have both been linked to long-run economic growth, and both exhibit imbalance in our sample. As a first robustness check, panel A of appendix table B5 reports estimates controlling for baseline ethnic fractionalization, urbanization, age structure, and education in equa-

tion (1).²⁵ The estimated decline in GDP due to splitting falls somewhat to 17.0% (S.E. = 7.6%). As a second robustness check, panel B presents estimates based on the trimmed sample of districts with an urbanization rate that lies within the common support (0 to 71%). The estimate is virtually unchanged, showing a GDP decline of 18.9% (S.E. = 8.0%). Finally, panel C presents estimates that use the trimmed sample and control for the aforementioned covariates. The estimate becomes larger in absolute value, indicating a GDP decline of 20.3% (7.9%). The conclusion that district splits reduce growth is quite robust to adjusting for covariate imbalance or imposing common support in the urbanization rate. (See appendix figure B2 for horizon-specific estimates.)

Alternative measures of economic activity. It is unusual to find a measure of GDP at the level of granularity available in Indonesia, especially over such a long period. As a result, much of the literature looking at subnational growth in developing countries has tended to rely on alternative measures like household expenditure and nighttime luminosity. We investigate the impact of splits on these measures of economic activity in detail in appendix B and find that they corroborate the impact on GDP. Splits result in a decline in household expenditure that is similar in magnitude to the decline in GDP (appendix figure B3). Nighttime luminosity also falls after a split, though the estimates are imprecise (appendix figure B3). We are reassured that our findings are not driven by measurement error in district GDP.

C. Early versus Late Splits

The baseline estimates are a weighted average of cohort-specific treatment effects, which could in principle vary across cohort. To examine this possibility while maintaining reasonable sample sizes, we estimate separate effects for

²⁴Crucially, the increase in spending that results from these transfers does not signal an increase in future taxes, which could lead to an increase in GDP through wealth effects.

²⁵Note that the model allows the impact of baseline covariates to vary by year and treatment cohort, but it does impose that the conditional means of potential outcomes are linear in the covariates (Wooldridge, 2025).

early (2002–2004) and late (2008–2009) splits, which are reported in table 1.

Early and late splits could have different impacts on growth for a few reasons. First, in the early period, district officials and voters had less experience with decentralized governance. All district governments, whether or not they split, had to contend with new expenditure and administration responsibilities, yet initially there was confusion over which level of government was responsible for specific functions. In this context, it would take time to establish chains of accountability between civil society, bureaucrats, and elected officials. Without these systems in place, the beneficial effects of district proliferation through increased accountability may have been impeded.

Second, because direct mayoral elections were rolled out over 2005 to 2008, all early splits occurred under indirectly elected mayors, and all late splits occurred under directly elected mayors. Consequently, any increase in yardstick competition due to splitting should be stronger for late splits. Late splits should therefore have more beneficial (or less harmful) effects than early splits if yardstick competition is an important mechanism.

Finally, the central government tightened the regulations on splitting after the first moratorium, increasing the minimum number of subdistricts that each parent and child district needed to have. Late splits may have resulted in less geographically fragmented service provision as a result.

Despite these differences, there are important shared features between early and late splits that could result in similar impacts. For instance, the challenge of recruiting competent civil servants to staff the newly formed districts was equally significant for both early and late splitters. (We discuss government personnel in section VIB.) If the fall in GDP is primarily driven by a deterioration in bureaucratic capacity, then early and late splits should have similar impacts.

Surprisingly, we find that late splits actually caused a larger GDP decline than early splits, though the difference between the two estimates is not statistically significant (table 1). Appendix figure B5 plots the dynamic effects of early and late splits. Due to the timing of these splits, we are able to estimate the impact of early splits on growth over a longer time horizon, and we can test for common trends for late splitters over a longer preperiod. The GDP dynamics in the first five years following a split are similar for early and late splitters. (For early splitters, GDP continues to fall in years 5 and 6 following the split, after which time the estimates become quite imprecise.) Late splitters do exhibit a downward—albeit imprecisely estimated—pretrend relative to nonsplitters; however, the differential pretrend is absent when we condition on baseline district covariates (appendix figure B6).

Both early and late splits appear to reduce GDP, but the relative magnitudes vary depending on the covariates and sample that we use, as shown in appendix table B5. In the trimmed sample (panel B), the estimates for early and late splits are quite similar. However, in the full sample with added controls (panel A), late splits have a more pronounced

negative impact on GDP, while early splits show a stronger negative effect in the trimmed sample with additional controls (panel C). This sensitivity could be attributed to the smaller sample sizes for each subgroup of splitting districts. As a consequence, we cannot draw firm conclusions about the nature of treatment effect heterogeneity across splitting cohorts. (See appendix figures B6 to B8 for the horizon-specific estimates.)

D. Spillover Effects and Aggregate Growth Impact

To what extent are the results informative for the aggregate effects of district splitting? The baseline estimates could underestimate the aggregate impacts if district splits have negative spillover effects on nearby districts. Alternatively, the estimates could overstate the negative effects of splitting if splits cause firms to simply relocate their activity from splitting districts to nonsplitting districts, leading to positive spillover effects.

One way to evaluate spillover effects is to estimate the impact of fragmentation at increasingly higher levels of geographic aggregation. We do this by estimating the following equation,

$$\frac{Y_{j,t+h} - Y_{j,t-1}}{Y_{j,t-1}} = \beta_h (\ln N_{j,t} - \ln N_{j,t-1}) + \lambda_{r(j),t,h} + \varepsilon_{j,t,h}, \quad (3)$$

where $N_{j,t}$ is the number of districts contained in geographic unit j in year t , and the $\lambda_{r(j),t,h}$ are region-by-year effects. The geographic units considered are original district (2000 borders), the intersection of province and island, and province. The advantage of considering province-by-island units is that they are, by construction, more numerous than provinces, increasing statistical precision. Furthermore, spillovers across islands may be limited.

The functional form of equation (3) assumes constant effects of proportionate changes in the number of districts. For example, increasing the number of districts from one to two is assumed to have the same impact on growth as increasing the number of districts from two to four. Hatfield and Kosec (2013) make the same functional form assumption in studying the effect of the number of county governments on local income growth in the United States. It is unfortunately necessary to make parametric assumptions in order to compare effects at different levels of aggregation. The effect of doubling the number of districts in year t on the outcome in year $t + h$ is $\ln(2) \cdot \beta_h$.

Table 2 reports the cumulative effect of doubling the number of districts, $\ln(2) \cdot \sum_{h=0}^4 \beta_h$.²⁶ Doubling the number of districts within the original 2000 borders leads to a cumulative decline in GDP by 13.4% (S.E. = 7.1%). This is somewhat smaller than the weighted CATT estimate of 18.5%. One potential reason is the functional form

²⁶ Appendix figure B9 plots the dynamic effects, and appendix table B.6 presents analogous results for nighttime luminosity.

TABLE 2.—GOVERNMENT FRAGMENTATION AND ECONOMIC GROWTH AT DIFFERENT LEVELS OF AGGREGATION

	Level of Aggregation		
	(1) Original District	(2) Province by Island	(3) Province
Cumulative Effect of Doubling Number of Districts	−0.134* (0.071)	−0.177* (0.094)	−0.419* (0.217)
Observations	2,979	504	297
Clusters	331	56	33

This table reports estimates of the cumulative effect of doubling the number of districts, $\ln(2) \cdot \sum_{h=0}^4 \beta_h$, obtained by replacing the outcome in equation (3) with $\sum_{h=0}^4 (Y_{d,t+h} - Y_{d,t-1}) / Y_{d,t-1}$ and scaling the coefficient by $\ln(2)$. The model is estimated at three different levels of aggregation, as indicated. Standard errors, reported in parentheses, are robust to heteroskedasticity and clustering by geographic unit. * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

assumption, as the CATT estimator is nonparametric. Another potential explanation is that the first split slightly more than doubles the number of districts within the original borders, yielding an average of 2.28 districts.²⁷ Scaling the estimate by $\ln(2.28)$ rather than $\ln(2)$ produces a cumulative decline of 15.9%, which is closer to the CATT estimate. The impact of doubling the number of districts at the province-by-island level is larger, showing a cumulative GDP decline of 17.7% (S.E. = 9.4%). The province-level estimate is larger still, showing a decline of 41.9% (S.E. = 21.7%).

The larger negative effects at higher levels of geographic aggregation suggest that splits have negative spillover effects. However, this conclusion is only tentative, as the estimates in table 2 are imprecise—especially those at the province level. We can, however, rule out the possibility that splits simply result in a reshuffling of economic activity across space without having an aggregate impact.

We next investigate channels through which fragmentation might have had this negative growth impact. We find little evidence that the increase in number of jurisdictions led to productive competition among districts to increase accountability, reduce corruption, or improve the business environment in other ways. Instead, we find that the lower quality of bureaucrats staffing the new district contributed to the negative effects of fragmentation. We also find some evidence of economies of scale in the provision of public goods. Following a split, district expenditure skews heavily toward administrative expenditure due to the requirements of setting up a new government. Despite the large increase in transfers, productivity-enhancing infrastructure does not improve.

VI. Mechanisms

In this section, we discuss evidence for key mechanisms through which fragmentation could theoretically impact economic growth. In appendix B, we use machine learning methods to further explore sources of treatment effect heterogeneity that could shed light on mechanisms.

²⁷Of the 98 districts that split for the first time in our sample period (excluding splits between 2012 and 2014), 76 divided into two districts, 17 divided into three districts, and five divided into four districts.

A. Economies of Scale

The creation of new governments entails greater spending on government administration, especially in the short run, as buildings are constructed and personnel hired. Large fixed costs of government administration, in turn, imply economies of scale in the overall provision of public services. This is true even if changing the size of districts does not change the scale of public service providers, such as schools and health clinics. In contrast to the bureaucratic capacity channel, economies of scale are relevant for both parent and child districts. We find that this loss of benefits from scale can be substantial.

Figure 3 shows that total expenditure steadily increases after a split, rising by 16% of pre-split GDP four years later. For context, district spending was about 12% of GDP in 2001. The figure also shows the dynamic impact of splits on the district expenditure shares in five functional categories: administration, human capital, physical capital, economic development, and social services. The share of expenditure devoted to administration sharply increases in the year of the split and remains at an elevated level four years later. The average increase over four years is 3.8 percentage points, which is a 12% increase relative to the baseline share of 32% (appendix table B7). This is achieved through a decrease in the share of spending devoted to human capital (education and health).

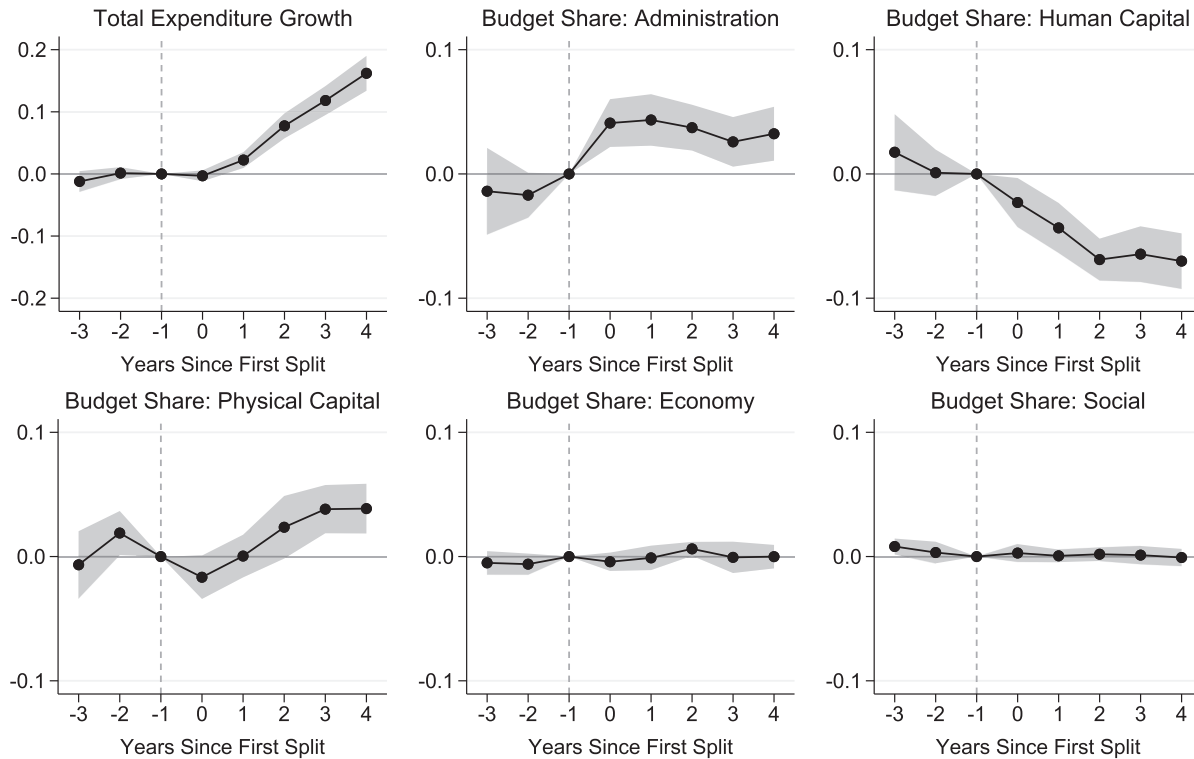
Some of these effects may change in the long run. For example, the share of expenditure on investment in physical capital initially falls but then starts to increase two years after the split. Overall, though, the spending composition results suggest a shift in spending toward public administration and away from productivity-enhancing areas.

Note that we cannot separate wages and salaries from other administrative expenditures. Part of the increase in administrative expenses could reflect rent seeking and patronage through civil service employment. Since half of the general grant allocation is used to finance wages, the central government regulates recruitment and employment contracts (Shah et al., 2012), making this less likely. Nevertheless, informal payments for entry and promotion within the civil service are common (World Bank, 2007). The increase in administrative expenses could therefore also suggest a worsening of accountability.

Despite the increased emphasis on administrative expenditure following splits, public services might still improve thanks to the large increase in fiscal transfers. Figure 4 displays the average effects of splitting on ten public services in the areas of education, health, public safety, and infrastructure. These data come from the Village Potential Statistics (Pendataan Potensi Desa, PODES).²⁸ We report separate estimates for original districts, parent districts, and child districts. For each public service, the estimates

²⁸Appendix figure B10 plots the dynamic effects for the PODES outcomes, while appendix figure B11 plots estimates for household outcomes taken from the National Socioeconomic Survey (Survei Sosial Ekonomi Nasional, SUSENAS).

FIGURE 3.—THE EFFECT OF DISTRICT SPLITS ON DISTRICT EXPENDITURE



This figure plots estimates of the cohort-size-weighted CATT (equation (2)) and their 95% confidence intervals. Total expenditure growth is measured as the change in total expenditure from year $t - 1$ to year $t + h$ relative to district GDP in year $t - 1$. The remaining outcomes are measured as the change in the expenditure share from year $t - 1$ to year $t + h$. The estimate for total expenditure at $h = -3$ is based on the 2003, 2004, 2008, and 2009 cohorts, because the data start in 2000. The estimates for $h \geq -2$ use all cohorts. Because the functional expenditures are only available for 2001–2012, the estimates for the budget share outcomes at $h = -3$ are based on the 2004, 2008, and 2009 cohorts; the estimates at $h = -2$ use the 2003, 2004, 2008, and 2009 cohorts; the estimates at $h \in \{0, 1, 2, 3\}$ use all cohorts; and the estimates at $h = 4$ use the 2002, 2003, 2004, and 2008 cohorts. All cohort weights are adjusted for the omission of specific cohorts. The confidence intervals are robust to heteroskedasticity and clustering by district.

indicate either no change or a decline following splits; the confidence intervals generally allow us to rule out moderate increases. These results are consistent with recent quasi-experimental evidence showing that decentralization worsens public service delivery in Vietnam (Malesky et al., 2014) and India (Chaudhary & Iyer, 2024). However, they contrast with the recent finding that reducing the size of local polities in Uttar Pradesh, India, improves the quality of public services (Narasimhan & Weaver, 2024).

The massive increase in spending induced by the general grant appears to have funded government administration without improving public service delivery. Economies of scale in governance may therefore help to explain the disappointing effects of splitting.

B. Bureaucratic Capacity

Another reason why fragmentation might harm GDP is the lack of bureaucratic capacity in newly created districts. There is no regulatory mechanism in place to reallocate civil servants from the parent district to the child district when a split occurs. Instead, civil servants transfer voluntarily, often due to geographical preferences or aspirations for career advancement. Most civil servants prefer to remain in the parent district with the established government, making it difficult for child districts to attract qualified bureaucrats. Civil ser-

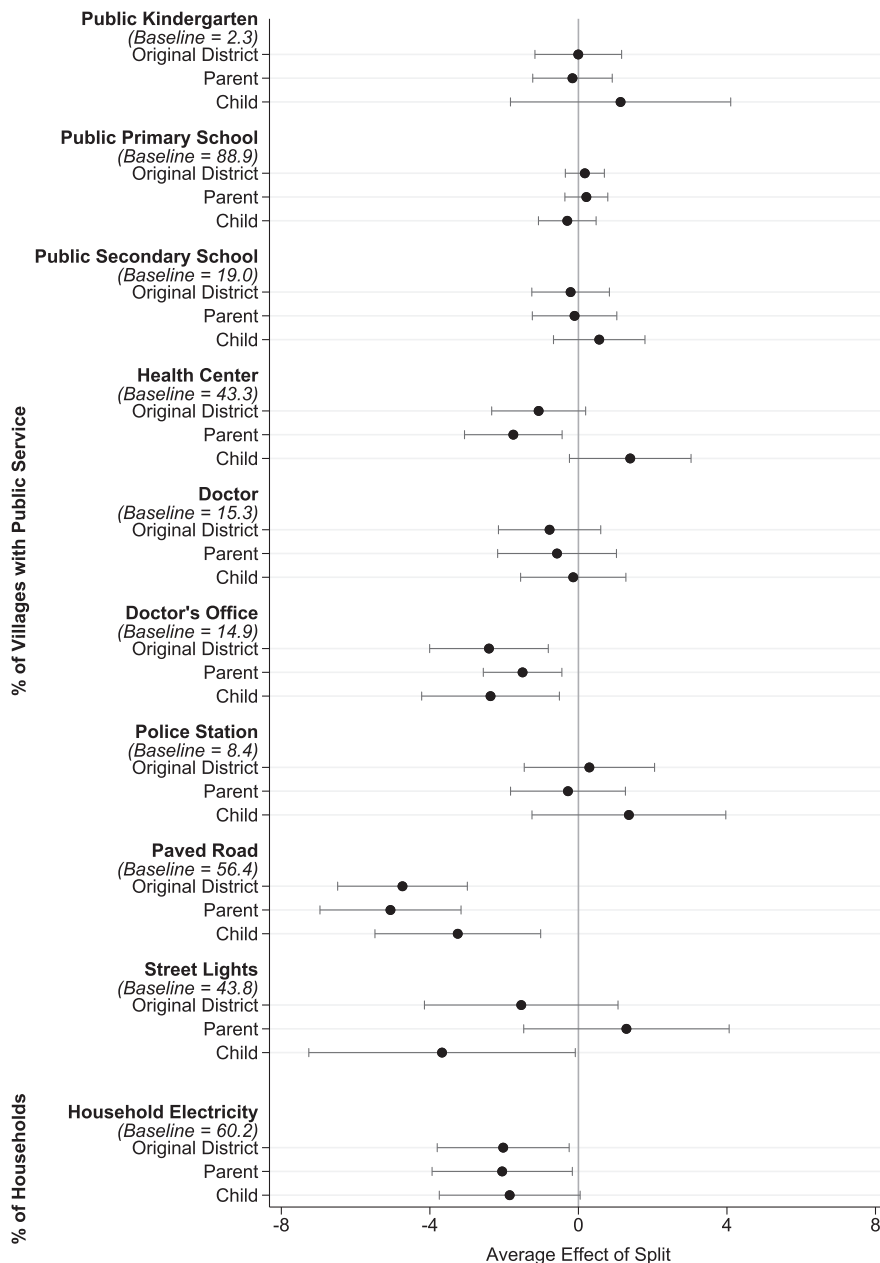
vants in child districts often have less experience but are promoted prematurely to meet regulatory requirements (Badan Perencanaan Pembangunan Nasional, 2007). Moreover, setting up a new government is a cumbersome process that disrupts the core activities of public administration.

Appendix table B8 shows that bureaucracies in child districts are initially understaffed compared to parent districts, although they catch up two to three years after the split (panel A). However, there are persistent differences in education and experience. Civil servants in child districts are less likely to hold a bachelor's degree than their counterparts in parent districts, potentially reflecting the challenge of recruiting high-quality candidates.²⁹ Child districts also tend to employ younger civil servants, which suggests they have less on-the-job experience (panels B and C).

Given these problems—and the importance of bureaucracy for development (Besley et al., 2022)—we expect the negative growth effects of splitting to be more pronounced for child districts. One challenge is to impute the increase in the general grant in parent and child districts separately due to the split, since the grant goes to the combined district prior to the split. We make the conservative assumption that the

²⁹By contrast, child districts have more educated mayors than parent districts, on average (appendix table B9). Attracting qualified mayoral candidates in the child district is not a problem, given the prominence of the position.

FIGURE 4.—THE EFFECT OF DISTRICT SPLITS ON ACCESS TO PUBLIC SERVICES (PODES)



This figure plots estimates of the average effect of the first district split and their 95% confidence intervals, based on the cohort-size-weighted CATT estimator (equation (2)). All outcome variables come from PODES. The first nine outcomes are the percentage of villages that have the public service, and the last outcome is the percentage of households with electricity. Each outcome is measured as the average change in the public service from the (three-year) period prior to the split to the two (three-year) periods after the split, $(1/2) \sum_{h=0}^2 (S_{d,e+h} - S_{d,e-1})$. The estimates use a balanced panel of 329 districts and control for region dummies, ethnic fractionalization, urbanization rate, share of population aged 15–64, share of population with a primary education, and share of population with a secondary education, all measured in 2000. The confidence intervals are robust to heteroskedasticity and clustering by district.

original district government would have spread expenditure across the district in proportion to the local population. A plausible alternative is that per capita expenditure before the split was higher in what becomes the parent district, as unequal treatment across regions may have motivated the split. In this case, we would be overestimating the increase in expenditure in parent districts and underestimating the increase in child districts.

Table 3 reports the cumulative effects of splitting separately for parent and child districts, while appendix fig-

ure B12 plots the dynamic effects. The general grant increases in both parent and child districts after the split, but the increase is three times as large in the child district. Nevertheless, parent and child districts experience similar cumulative declines in GDP.³⁰ Relative to predicted GDP based on

³⁰The sample size is smaller compared to table 1 because the GDP data are not always backcast to the year prior to the split for the parent and child districts. The cohort-size weights are adjusted to reflect the relative frequencies of each splitting cohort conditional on having nonmissing GDP growth rates.

TABLE 3.—GOVERNMENT FRAGMENTATION AND ECONOMIC GROWTH: PARENT VERSUS CHILD DISTRICTS

	General Grant	Predicted GDP		Actual GDP
	(1)	(2) Multiplier = 0.6	(3) Multiplier = 1.8	(4)
Panel A: Parent Districts				
Cumulative Effect of Split	0.196*** (0.021)	0.118*** (0.013)	0.354*** (0.038)	− 0.205** (0.085)
Observations	1,258	1,258	1,258	1,258
Districts	326	326	326	326
Panel B: Child Districts				
Cumulative Effect of Split	0.594*** (0.049)	0.357*** (0.029)	1.070*** (0.088)	− 0.138* (0.074)
Observations	1,258	1,258	1,258	1,258
Districts	326	326	326	326

This table reports estimates of the cumulative effect of the first district split, $\sum_{h=0}^4 \beta_h$, based on the cohort-size-weighted CATT estimator, equation (2). The outcome is measured as the cumulative growth from the year prior to the split to four years after the split, relative to GDP in the year prior to the split, $\sum_{h=0}^4 (Y_{d,e+h} - Y_{d,e-1}) / Y_{d,e-1}$. Separate estimates are reported for parent and child districts. The sample is restricted to districts for which the GDP data are reliably backcast for parent and child districts at least one year prior to the split. Standard errors, reported in parentheses, are robust to heteroskedasticity and clustering by district. * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

fiscal multipliers, the GDP decline is much larger for child districts. For instance, a fiscal multiplier of 1.8 predicts a cumulative GDP increase of 107.0% in child districts and 35.4% in parent districts.

As a final piece of evidence, we show in appendix B that the short-run reduction in physical capital expenditure following a split is due to delays in the execution of spending plans, not changes in spending priorities. These delays are concentrated in child districts, suggesting that the inexperience of civil servants in these districts played a role.

Together, the results suggest that bureaucratic capacity is an important mechanism behind the negative growth impacts of splitting. However, bureaucracy cannot explain why GDP also declines in parent districts, so other mechanisms must also play a role.

C. Accountability

An often-cited rationale in favor of more localized governance is that it can increase accountability through either competition between jurisdictions or decreased distance between decisionmakers and citizens. Because Indonesian districts mainly control expenditure decisions and not tax policy, we might expect competition for the tax base to matter less in this context. However, districts rely on firms for revenue through licenses and permits administered by the district, and through business tax revenue collected and shared by the central government. We examine the impact of government fragmentation on measures of accountability that are particularly relevant for firms—indirect taxes and fees and informal taxation or “gifts.” We then test for downstream impacts on regulated activities, productivity, and competition.

To study firm-level outcomes, we modify equation (1) by adding industry-by-year effects to control for industry-specific shocks. Let $j(i)$ be the four-digit industry classifica-

tion of firm i . The regression is

$$Y_{i,d,e+h} - Y_{i,d,e-1} = \alpha_{j(i),e,h} + \beta_{e,h} D_d(e) + X'_{d,e,h} \lambda_{e,h} + (D_d(e) \cdot \dot{X}_d(e))' \gamma_{e,h} + \varepsilon_{i,d,e,h}, \quad (4)$$

where Y is a firm outcome measured in either logs or levels. We aggregate the estimates of $\beta_{e,h}$ using equation (2). To examine competition, we use equation (4) with outcomes that are measured at the “market” (i.e., industry-by-district) level.

Table 4 shows that increasing the number of local governments does not have a disciplining impact on bribery.³¹ In fact, we find some evidence of the opposite—the probability that a manufacturing firm reports a “gift” payment rises by around 3 percentage points following a split. On the other hand, the average bribe rate—gift payments as a share of firm revenue—does not change. By contrast, the probability that the firm pays any indirect taxes or fees does not change, but the average tax rate falls by 0.11 percentage points. The small effect on formal taxes is unsurprising given that most taxes are administered centrally. (Firms do not separately report local and central taxes and fees in the IBS survey.)

As Burgess et al. (2012) showed, Indonesia’s district proliferation increased corruption in illegal logging, allowing cheaper access to timber, consistent with a model of bribes as a fee for service and district governments as monopolistically competitive entities. When the number of competitors increases, the service price (bribes) falls and the quantity sold increases. Turning to the latter, we find

³¹ Appendix figure B13 plots the dynamic effects. There is a sharp drop in 2000 in the probability that the firm pays any gifts and the probability that the firm pays any indirect taxes or fees (appendix figure B14). The drop is not due to changes in the questionnaire or sample but may be linked to nationwide changes at the time. By contrast, there is no trend break in firm revenue. Because we cannot determine whether the trend breaks in bribes and taxes represent a change in measurement or a change in the real variable, we exclude data on gifts and taxes prior to the year 2000.

TABLE 4.—IMPACT OF FRAGMENTATION ON GIFTS, TAXES, AND REGULATED ACTIVITIES

	Gifts		Indirect Taxes and Fees		Regulated Activities	
	(1) Paid Any	(2) % Revenue	(3) Paid Any	(4) % Revenue	(5) Any Exports	(6) Any Land Contracts
Avg. Effect of Split	0.032*** (0.010)	−0.005 (0.016)	0.001 (0.010)	−0.112** (0.057)	0.018* (0.010)	0.010* (0.006)
Baseline Mean	0.623	0.303	0.720	0.974	0.181	0.056
Observations	76,324	75,774	76,756	76,151	54,064	76,324
Districts	307	307	308	308	294	307

This table reports estimates of the average effect of the first district split, based on the cohort-size-weighted CATT estimator—equations (4) and (2). The estimates are based on plant-level data. The outcomes are measured as the average change over the first five years following the split relative to the year prior to the split, $(1/6) \sum_{h=0}^5 (Y_{i,d,e+h} - Y_{i,d,e-1})$. Standard errors, reported in parentheses, are robust to heteroskedasticity and clustering by district. * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

some evidence that exporting and land contracts, which both require licenses, increase following splits (columns 5 and 6 of table 4).

The evidence presented so far suggests that intergovernmental competition modestly increases service access for firms while increasing the incidence of bribery. Overall, we do not find dramatic improvements in the business environment for manufacturing firms. We provide further evidence on the lack of improvement in government accountability and services using data from the Economic Governance Survey conducted by KPPOD (Regional Autonomy Watch) and the Asia Foundation. The survey was designed to measure how local governance affects the business environment across all parts of Indonesia. It contains a rich array of variables measuring the formal and informal costs of business licensing and corruption, as well as property rights enforcement. Two waves of the survey, in 2007 and 2011, collected data from two essentially nonoverlapping sets of districts, which together represented nearly all districts at the time.

Appendix figure B15 summarizes the difference-in-differences estimates for 48 outcome variables from the Economic Governance Survey.³² Each outcome is standardized to have a mean of 0 and a standard deviation of 1. The cost and wait variables are also winsorized at the 99% level to reduce the influence of a few extreme outliers. Most of the estimates are close to 0 and statistically insignificant. Six estimates are significant at the 5% level, though we would expect two or three estimates to be significant by random chance under the null of no effect, given the large number of outcomes considered. Overall, the results suggest that splitting has a small impact on the business environment. If anything, splitting leads to a reduction in business formality and an increase in the licensing burden. One caveat is that due to the timing of the surveys, these results represent short-term effects of late splits.

³²The regression specification is

$$Y_{f,d,t} = \beta_0 + \beta_1 \text{Splitter}_d + \beta_2 \text{Post}_t + \beta_3 \text{Splitter}_d \times \text{Post}_t + \lambda_{r(d),t} + \varepsilon_{f,d,t},$$

where $Y_{f,d,t}$ is a firm outcome, Splitter_d is an indicator variable that equals 1 for districts that split between 2008 and 2010, Post_t is an indicator variable that equals 1 if $t = 2011$, and $\lambda_{r(d),t}$ is a region-by-year effect. Districts that never split over the analysis period are used as controls.

The Economic Governance Survey contains large and small firms, while our manufacturing survey only covers firms with at least twenty employees. In the appendix, we present separate results from the Economic Governance Survey for firms with twenty or more employees (appendix figure B16) and for firms with fewer than twenty employees (appendix figure B17). The results suggest that the licensing burden increased for both large and small firms, but the reduction in formality was concentrated among small firms.

While splits do not generate significant measured improvements in the business environment, they could still generate unmeasured improvements, which should increase firm productivity or market competition. However, we find that splits do not significantly affect labor productivity, total factor productivity (TFP), or different measures of market competition (appendix table B12).³³ This is not to suggest that political economy factors do not matter for productivity in Indonesia. Abeberese et al. (2023) find that the introduction of democratically elected district mayors increased the productivity of these same manufacturing firms. District splits apparently do not generate enough intergovernmental competition to produce a similar effect.

VII. Discussion

The empirical literature on fragmentation and growth has found divergent results depending on the context. Even limiting our attention to seemingly similar contexts of large, developing economies in recent decades, we find differing experiences. Dahis and Szerman (forthcoming) find that splits of Brazilian municipalities increase nighttime luminosity in child districts and leave parent districts unaffected.

³³Appendix figure B18 plots the corresponding dynamic effects. We estimate (log) TFP following the approach of Ackberg et al. (2015), using total intermediate input costs (electricity, fuel, and raw materials) as a proxy for firm productivity. We implement their estimator using the `acfest` package in Stata (Manjón & Mañez, 2016). The firm's capital stock is not reported in IBS in 1996 and 2006, and it is missing for many firms in other years. We linearly interpolate missing capital values using values in adjacent years. Furthermore, we drop a few implausibly large values of capital and drop firms that exhibit large spikes in the capital variable. Specifically, we drop all values for firms that ever experienced a one- or two-year change in capital (in percentage terms) that was below the 1st percentile or above the 99th percentile.

Cohen (2024) finds that local government fragmentation in Uganda resulted in a decrease in nighttime luminosity, particularly in child districts.

Evaluating the source of heterogeneity in results is challenging given the complexity of relevant differences. For example, the empirical strategies that these papers pursue mean that the results are necessarily local to the size of the jurisdictions that split. As Narasimhan and Weaver (2024) show, fixing all other characteristics of administration such as the level of fiscal autonomy, experience of bureaucrats, and economies of scale, population size in the local administrative unit can affect economic outcomes.

In Brazil, the authors attribute the result to the expansion of the public sector, as they find no impacts on the private sector. Since annual GDP data are not available for Brazilian municipalities, it is not possible to directly compare their estimates with ours. This is especially important because one possible explanation for the difference in results is that the increase in transfers due to splitting is larger relative to the local economy in Brazil than in Indonesia—something we cannot evaluate without knowing local GDP. In Uganda, central government transfers to districts do not change very much, but there are substantial reductions in foreign aid per capita, particularly in newly created districts, mirroring the decline in growth.

A common theme in all settings seems to be that government capacity matters. New district governments with new officials fare worse. In Indonesia, even parent districts were inexperienced with expenditure responsibilities; prior to decentralization in 2001, administration was heavily centralized. This is unlike both the Brazilian and Ugandan contexts where local governments in the parent district retained an electoral setting and administrative responsibilities that had been in place for many years. Further work in this area could more directly explore the importance of the experience of local government officials, especially in the administrative duties with which they are tasked.

An open question is under what conditions decentralization can improve accountability. In Brazil and Indonesia, there is evidence of greater rent seeking and little increase in accountability (de Andrade Lima & Neto, 2018). This is surprising because greater accountability is an often-cited rationale for decentralization—and, as Narasimhan and Weaver (2024) show, reducing the polity size can result in greater accountability through electoral mechanisms. A notable contrast here again is the amount of experience both citizens and elected officials have with existing institutional structures. In the Indian context, a village joins a smaller or larger existing district, leaving electoral structures and administrative responsibilities unchanged.

Our results also appear to be at odds with evidence from the United States and China. Stansel (2005) and Hatfield and Kosec (2013) find that greater local government fragmentation correlates with faster economic growth in the cross section of US metropolitan areas. Their context differs from ours: the United States is a mature democracy with greater

decentralization of tax-setting authority. Beyond differences in context, these findings capture longer-term impacts and relate to governments that have existed for a long time. Our estimates, by contrast, capture short-run effects for very young governments. China presents a unique context in which the central government provides explicit incentives for subnational governments to compete with one another and promote growth. In this context, Zhang et al. (2019) find that the fragmentation of municipal governments promotes growth to an extent. Increasing the number of districts from one to two increases growth, but further increases reduce growth on the margin. Similar to the studies on the United States, Zhang et al. (2019) exploit cross-sectional variation and study governments that have existed for a while.

Several factors could contribute to the positive impacts of fragmentation found in the United States and China. First, the negative impact of reduced bureaucratic capacity may not be present, given that the local governments in these studies had existed for a long time. Hatfield and Kosec (2013) note that “out of over 3000 counties in the continental United States, only six new functional counties have formed since 1970, and only six functional county governments have dissolved since 1970.” Second, incentives for intergovernmental competition may have been stronger, given differences in the institutional environment. Nonetheless, the nonmonotonic effect of fragmentation in Zhang et al. (2019) supports the idea that fragmentation can harm growth due to economies of scale.

VIII. Conclusion

The fragmentation of districts in Indonesia represents a major exercise in decentralization in a populous country. Exploiting idiosyncratic variation in the timing of district splits, we find that fragmentation reduces GDP in the short run. The results are robust to flexibly controlling for region effects and baseline levels of ethnic fractionalization, urbanization, age structure, and education. Bureaucratic capacity and economies of scale are important mechanisms that explain the negative impact of fragmentation. While splitting could increase accountability via stiffer intergovernmental competition, this benefit does not materialize in practice—at least in the short run.

An open question is how fragmentation will affect long-run growth. Because the district GDP series ends in 2013, we focus on the GDP impact up to four years after a split. For early splitters, we show that GDP continues to fall in years 5 and 6. Beyond that, the estimates are too imprecise to draw firm conclusions. Given that bureaucratic capacity appears to be an important mechanism, fragmentation should be less harmful for GDP in the long run as the new civil servants gain experience and recruiting improves as the government becomes more established. The long-run effects should also depend on the extent to which the increase in administrative expenditure reflects one-time spending

on new government infrastructure rather than permanent increases in government payroll.

Government fragmentation remains an active process throughout the world.³⁴ Future research should examine the long-run effects of fragmentation in other settings. An important question is how to limit the disruption caused by the creation of new local governments, so that localities can better enjoy the potential benefits of decentralization.

³⁴ As of July 2017, there were 246 pending applications to create new provinces and districts in Indonesia (Tempo, 2017).

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